EXPERIMENT 3

EDA – DATA CLEANING

AIM

To perform data cleaning using Pandas by identifying missing values, removing duplicate records, correcting data types, and checking data consistency before analysis.

THEORY

Data cleaning is an important step in exploratory data analysis. Real-world datasets may contain missing values, duplicate rows, inconsistent category names, incorrect data types, or values that fall outside expected ranges. These issues can affect summary statistics and visualizations. Pandas provides methods such as isnull(), dropna(), fillna(), duplicated(), drop_duplicates(), astype() and replace() to inspect and correct common data-quality problems.

ALGORITHM / PROCEDURE

1. Import Pandas and load the dataset into a DataFrame.
2. Display the first few rows and inspect the data types.
3. Count missing values in each column.
4. Identify duplicate rows and remove exact duplicates if appropriate.
5. Handle missing numeric values using a suitable method such as median imputation, or remove rows when justified.
6. Standardize inconsistent text labels and convert columns to the correct data types.
7. Recheck missing values, duplicate rows and data types after cleaning.
8. Save the cleaned dataset to a new file.

SOURCE CODE

```
import pandas as pd

# Load the dataset
df = pd.read_excel("socialmedia_dataset.xlsx")

print("Original shape:", df.shape)
print("\nMissing values:")
print(df.isnull().sum())

# Remove exact duplicate rows
df = df.drop_duplicates()

# Standardize text columns
for col in df.select_dtypes(include="object").columns:
    df[col] = df[col].str.strip()

# Fill missing numeric values with the median
numeric_cols = df.select_dtypes(include="number").columns
for col in numeric_cols:
    df[col] = df[col].fillna(df[col].median())

# Fill missing categorical values with the mode
categorical_cols = df.select_dtypes(include="object").columns
for col in categorical_cols:
    if df[col].isnull().any() and not df[col].mode().empty:
        df[col] = df[col].fillna(df[col].mode()[0])

print("\nCleaned shape:", df.shape)
print("Remaining missing values:", df.isnull().sum().sum())
print("Remaining duplicate rows:", df.duplicated().sum())

# Save the cleaned dataset
df.to_excel("socialmedia_cleaned.xlsx", index=False)
print("Cleaned file saved successfully.")
```

OUTPUT

```
Original shape: (1000, 31)
Missing values: 0
Cleaned shape: (1000, 31)
Remaining missing values: 0
Remaining duplicate rows: 0
Cleaned file saved successfully.
```

The output shown is an example of the expected format. The actual values depend on the dataset used. If the input data has no missing values or duplicate rows, the corresponding cleaning steps may not change the number of rows.

RESULT

The dataset was inspected and cleaned using Pandas. Duplicate rows were removed, missing values were handled where present, text values were standardized, and the cleaned data was saved to a new Excel file for further analysis.